half a million to a million lives. Among those granted life would have been my brother Joe, killed in October 1944 in the Battle for Italy. What a difference it would have made if the critical date [of the atomic bomb's first use in the war] had been not August 6, 1945, but August 6, 1943."

In the U.S.S.R., physicists abandoned all nuclear research in June 1941, when Germany attacked Russia, since other physics would produce quicker payoffs for national defense. As the German army marched on and surrounded Leningrad, Zel'dovich and his friend Khariton were evacuated to Kazan, where they worked intensely on the theory of the explosion of ordinary types of bombs, trying to improve the bombs' explosive power. Then, in 1943, they were summoned to Moscow. It had become clear, they were told, that both the Americans and the Germans were mounting efforts to construct an atomic bomb. They were to be part of a small, elite, Soviet bomb development effort under the leadership of Igor V. Kurchatov.

By two years later, when the Americans bombed Hiroshima and Nagasaki, Kurchatov's team had developed a thorough theoretical understanding of nuclear reactors for making plutonium-239, and had developed several possible bomb designs—and Khariton and Zel'-dovich had become the lead theorists on the project.

When Stalin learned of the American atomic bomb explosions, he angrily berated Kurchatov for the Soviet team's slowness. Kurchatov defended his team: Amidst the war's devastation, and with its limited resources, the team could not move more rapidly. Stalin told him angrily that if a child doesn't cry, its mother can't know what it needs. Ask for anything you need, he commanded, nothing will be refused; and he then demanded a no-holds-barred, crash project to construct the bomb, a project under the ultimate authority of Lavrenty Pavlovich Beria, the fearsome head of the secret police.

The magnitude of the effort that Beria mounted is hard to imagine. He commandeered the forced labor of millions of Soviet citizens from Stalin's prison camps. These *zeks*, as they were colloquially called, constructed uranium mines, uranium purification factories, nuclear reactors, theoretical research centers, weapons test centers, and self-contained, small cities to support these facilities. The facilities, scattered across the face of the nation, were surrounded by levels of security unheard of in the Americans' Manhattan Project. Zel'dovich and Khariton were moved to one of these